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Rathin Singha

I am a fifth-year PhD student at UCLA (graduating Spring 2027), co-advised by Todd Millstein and George Varghese, seeking full-time research scientist roles. My research interests are computer networks and systems, network verification, testing, and building agentic AI pipelines. More specifically, I work on network configuration analysis and incremental control-plane simulation (fast what-if analysis of configuration changes and failures in Batfish), testing network implementations using generative AI and symbolic execution, and synthesizing verifiably correct and robust implementations using modularity techniques.

Education

2022–2027 **Ph.D (expected Spring 2027)**, *Computer Science, UCLA [***GPA: 3.87/4.0***]*

2021–2022 **MS**, *Computer Science, UCLA [***GPA: 4.0/4.0***]*

2017–2021 **B.Tech (Honors)**, *Computer Science and Engineering, IIT Kharagpur [***GPA: 9.32/10***]*

Publications

NSDI'26 Rajdeep Mondal*, **Rathin Singha***, Todd Millstein, Ryan Beckett, Siva Kesava Reddy Kakarla, George Varghese. *Eywa: Automating Model-Based Testing using LLMs.* USENIX NSDI 2026. (*Joint first authors)

arXiv'26 **Rathin Singha**, Kuan Qian, Srinath Saikrishnan, Tracy Zhao, Soheil Abbasloo, Ryan Beckett, Siva Kesava Reddy Kakarla, Todd Millstein, George Varghese. *CornerCase: Automated Extremal Testing of Protocol Implementations using LLMs.* arXiv:2606.29124, 2026.

HotNets'24 **Rathin Singha**, Rishabh Iyer, Charles Liu, Caleb Terrill, Todd Millstein, Scott Shenker, George Varghese. *If Layering is useful, why not Sublayering?* ACM HotNets 2024.

NSDI'24 **Rathin Singha**, Rajdeep Mondal, Ryan Beckett, Siva Kesava Reddy Kakarla, Todd Millstein, George Varghese. *MESSI: Behavioral Testing of BGP Implementations.* USENIX NSDI 2024.

EMNLP'22 Jivnesh Sandhan, **Rathin Singha**, Narein Rao, Suvendu Samanta, Laxmidhar Behera, Pawan Goyal. *TransLIST: A Transformer-Based Linguistically Informed Sanskrit Tokenizer.* Findings of EMNLP 2022.

Research Projects

Jul'26–Present **Incremental Batfish: Faster simulation of Network Configuration Changes [To be submitted to SIGCOMM'27]**
[*Mentors: Todd Millstein, George Varghese, Dan Halperin (Amazon)*]
Extended Batfish to recompute only what a configuration change affects, rather than re-simulating the whole network. Covers BGP policy edits, link failures and recoveries, and device additions and removals, with incremental OSPF repair, prefix-level forwarding-table updates, and reuse of BGP session checks. On a 1,001-router Internet2-based benchmark it gives 16–19× speedups for policy edits and 3.8–5.1× for link changes, matching Batfish's output exactly. Speedups grow with network size. On 276 real ISP topologies from the Internet Topology Zoo, speedups again grow with size, reaching 7–25× for policy edits on the largest network (754 routers).

Jun'26–Present **AutoDesign: LLM-Driven Design Automation for Verifiable Problems [To be submitted]**
[*Mentor: Soheil Abbasloo*]
Built a multi-agent LLM system (Analyst, Architect, Implementer, Diagnostician) that, given only a problem and an automatically checkable success criterion, designs a solution pipeline, implements and runs it, grades the result, and repairs its own design until the criterion is met. On Mahak's congestion-control performance-mapping task it met the accuracy goal within four self-correcting iterations, by a different route than the original authors (no active learning). The same loop, unchanged, designed a KLEE-based DNS test generator for Eywa, producing 1,554 harness-valid CNAME tests that structurally match the triggers of 5 of 6 known bugs, with no human modeling effort.

Jul'25–Present **CornerCase: LLM-Driven Extremal Testing of Network Protocols [To be submitted to SIGCOMM'27]**
[*Mentors: Todd Millstein, George Varghese, Soheil Abbasloo*]
Designed an LLM-based pipeline that reads RFCs, extracts validity constraints, and generates boundary-focused ("extremal") test suites for protocols such as DNS, BGP, SMTP, HTTP, QUIC, and shortest-path libraries; differential testing across implementations has already revealed numerous semantic inconsistencies and previously unknown bugs.

Jun'25–Sep'25 **Autonomous Multi-Agent LLM Test-Suite Generation for Internet Protocols (Research Intern, Microsoft Research Redmond)**
[*Mentor: Soheil Abbasloo*]
Built a dialectical multi-agent framework where LLMs iteratively propose, critique, and refine protocol tests. On BGP and DNS benchmarks, it achieved 95–100% test-quality within 2–3 cycles across GPT-4.1, GPT-4o, and o1—where single-agent baselines plateaued below 80%. Work is being extended to automate assessment frameworks for congestion control algorithms.

Mar'24-Sept'25 **Eywa: Automating Model Based Testing using LLMs [NSDI 2026, Joint First Author] [Link]**
[*Mentors: Ryan Beckett, Siva Kakarla, Todd Millstein, George Varghese*]
Created an LLM-based tool to automatically generate models for network protocols like DNS and BGP; surprisingly, LLM errors enrich the test suite rather than affect coverage. We found 15 new bugs in 10 DNS implementations and 4 new bugs in 3 BGP implementations.

June'24-Sept'24 **Microsoft Research Internship (Bangalore): Generating Models from Specifications using LLMs**
[*Mentor: Akash Lal*]
Developed a tool (MockGen) using an LLM to generate models (called mock implementations) as well as high-coverage test cases from OpenAPI specifications for Azure services such as DNS and BlockBlob. Unlike Eywa, which uses symbolic execution, I used the LLM to *directly* generate tests and *fed back* the test results to the LLM to improve the mock. This achieved a mock accuracy of 95% relative to manually written mocks.

Sept'23-May'24 **If Layering is useful, why not sublayering? [HotNets 2024, Presented Paper] [Link]**
[*Mentors: Todd Millstein, Scott Shenker, George Varghese*]
Proposed recursively breaking up monolithic layers into simpler sublayers for ease of verification, debugging and principled hardware offload. Proposed a new sublayering for TCP and showed sublayering could simplify verification for HDLC bit-stuffing using Coq.

June'23-Sept'23 **Microsoft Research Internship (Redmond): Securing Azure RBAC with Static Analysis [Applied for Patent]**
[*Mentors: Behnaz Arzani, Ryan Beckett, Nikolaj Bjorner, Siva Kakarla*]
Developed a static analysis tool for Azure RBAC to verify policies, detect duplicates, and generate exhaustive tests for Azure Implementations. We reduced the verification time for large policies from 30 minutes to 0.16s using various optimization algorithms. We also found 5 discrepancies across C# and C++ implementations of Azure RBAC.

April'22-June'23 **MESSI: Behavioral Testing of BGP Implementations [NSDI 2024, Presented paper] [Link]**
[*Mentors: Ryan Beckett, Siva Kakarla, Todd Millstein, George Varghese*]
Automated testing of BGP Implementations such as Quagga, FRR, GoBGP, Batfish, by generating high coverage tests using symbolic execution and enumerative testing. Solved BGP-specific challenges due to statefulness, regular expressions, and incremental updates. Our tool found 22 bugs across 5 BGP implementations.

April'21-Sep'21 **TransLIST : A Transformer-Based Linguistically Informed Sanskrit Tokenizer [Findings of EMNLP 2022] [Link]**
[*Mentors: Laxmidhar Behera, Pawan Goyal, IIT Kanpur*]
Developed a novel Sanskrit word segmentation model combining character-level input with latent word information and an ensemble-based constrained inference algorithm, achieving 1.8-point F-score and 7.2-point PM gains over SOTA.

Oct'21-March'22 **Automated Verification of Network Protocol Implementations using NLP**
[*Mentors: Todd Millstein, George Varghese, UCLA*]
Built a model that automatically generates constraints from the RFC texts of protocols like ICMP, IGMP, etc by using Natural Language processing. These constraints can be used to generate test cases to check RFC compliance with the existing implementations.

Other Internships

June'22-Sept'22 **Intel Corporation : Auto Thresholding of Transport Traffic in MsQuic Library**
[*Mentor: Rong Pan, now VP at AMD*]
Implemented a novel algorithm in MsQuic to dynamically expedite mission-critical flows based on congestion, achieving 3x and 10x latency improvements for two Google Data Center workloads.

April'20-August'20 **DAAD-WISE 2020 : Semantic Frame Identification**
[*Mentor: Chris Biemann, University of Hamburg*]
Enhanced semantic frame disambiguation using a fine-tuned BERT model, achieving a 1.2-point F-score improvement over SOTA on FrameNet data.

Technical Skills

Programming	Python, C++, C, Java, C#, Coq, Dafny
Networking	BGP, OSPF, DNS, TCP, QUIC, SMTP, HTTP
Verification	Batfish, KLEE, Z3, symbolic execution, differential testing
AI/ML	LLM multi-agent pipelines, PyTorch, TensorFlow, Keras, scikit-learn, NumPy, pandas
Tools	Docker, Mininet, Mahimahi, Wireshark

Coursework

Cloud Computing, Network Verification, Machine Learning, Advanced Machine Learning, Natural Language Processing, Internet of Things, Automated Reasoning, Bayesian Networks, Network Algorithmics

Awards

- UCLA Division of Graduate Education Fellowship Award for excellence in research
- DAAD-WISE 2020 Fellowship for a research internship in Germany
- LER Fellowship Award for being the 2nd highest rank to join IIT Kharagpur, with All India Rank of 178 in JEE Advanced
- Gold Medal in Silverzone International Mathematics Olympiad in 8th Standard